

14. MATHEMATICS AND MANAGEMENT SCIENCE

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ABSTRACT

Mathematics reveals hidden patterns that help us to understand the world around us. Now, much more than arithmetic and geometry, mathematics today is a diverse discipline that deals with data, measurements and observations from science, with inference, deduction, and proof; and with mathematical models of natural phenomena, of human behaviour, and of social systems. People in business, economics and the social sciences are increasingly aware of the need to be able to handle a range of mathematical tools. Mathematics is an important tool which all good managers should appreciate. The management science is designed to fill this need by extending in Mathematics and Statistics into several even more practical and powerful areas of mathematics. Such things as differential equations and stochastic processes, for example, do have direct, frequent and practical applications to everyday management situations. The course therefore both extends and reinforces existing knowledge and introduces new areas of interest and applications of mathematics in the ever-widening field of management.

Keywords : Mathematic, Diverse Discipline, Social Sciences, Stochastic Processes, field

INTRODUCTION

Importance of mathematics can be understood by the definition given by Galileo. He defined mathematics as '*a language in which God has written the world*'. '*Mathematics is the queen of Sciences.*' Those are the words of Carl Frederick Gauss. Much the same sentiment is expressed in the ancient Sanskrit verse:

यथा शिक्षा मयूराणां नागानां मणयो यथा ।

तथा वेदाङ्गशास्त्राणां गणितं मूर्धनि स्थितम् ॥

Roger Bacon (1214-1294), an English Franciscan friar, philosopher, scientist and scholar of the 13th century, once stated:

"Neglect of mathematics works injury to all knowledge, since he who is ignorant of it cannot know the other sciences or the things of the world."

'What is mathematics?' The answer to this question is of course complex; there are elaborate elucidations, some excellent, on the subject but inevitably, even the best accounts give incomplete answers.

Mathematics is a branch of science, which deals with numbers and their operations. It involves calculation, computation, solving of problems etc. Its dictionary meaning states that, 'Mathematics is the science of numbers and space' or 'Mathematics is the science of measurement, quantity and magnitude'. It is exact, precise, systematic and a logical subject.

IMPORTANCE OF MATHEMATICS

The literal meaning of mathematics is "things which can be counted" now you can think that counting has vital role in our daily life; just imagine that there were no mathematics at all, how would it be possible for us to count members of the family, number of students in the class,